

STA301

Statistics & Probability

Final Term Most Important & Repeated Mcqs

❖ **Midterm Short Notes**

❖ **Final Term Short Notes**

❖ **Mid Term Past Paper**

❖ **Final Term Past Paper**

❖ **Mega File**

❖ **Important Question**

❖ **Repeated Question**

❖ **Solved Past Paper**

❖ **Moaz & Waqar**

❖ **Accurate Assignment**

❖ **Quiz File**

❖ **Accurate GDB**

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Which is not a source of existing statistics?

- Government
- International agencies
- **Personal interviews**
- Private sources

Followings are the advantages of secondary data except

- Non-Reactivity
- **Selective survival**
- Low cost
- Spontaneity

Following are the disadvantages of secondary data except

- **Longitudinal analysis**
- Sampling bias
- Coding difficult
- Incompleteness

The coding of the secondary data may be difficult because of

- Differences in content or subject matter
- Lack of standardization
- Differences in length and format
- **All of these**

The researcher must be concerned about the following problems while using secondary data in research.

- Validity
- Reliability
- **All of these**
- None of these

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If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► 0,1,2,3 rep
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

When c is a constant, then $E(c)$ is:

- ► 0
- ► 1
- ► c
- ► $-c$

When the random variable X and Y are independent, its co-variance is:

- ► One
- ► Negative
- ► Zero
- ► Positive

If X and Y are independent variables, then $E(XY)$ is:

- ► $E(XY)$
- ► $E(X) \cdot E(Y)$
- ► $X \cdot E(Y)$
- ► $Y \cdot E(X)$

Which dispersion is calculated from all the observations?

- ► Standard deviation
- ► Quartile deviation
- ► Rang
- ► Coefficient of Rang

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Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

- ► 49
- ► Sqrt (7)
- ► 0
- ► 7

Which one is the poor measure of dispersion in open-end distribution?

- ► Range
- ► Standard deviation
- ► Mean deviation
- ► Variance

Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct:

- ► The correlation is -5
- ► The correlation is 5
- ► The correlation is 1
- ► The correlation is 0

Sum of absolute deviations of the values is least when deviations are taken from:

- ► Mean
- ► Median
- ► Mode
- ► Geometric mean

Which of the following measures of central location is affected most by extreme values:

- ► Geometric Mean
- ► Median
- ► Mean
- ► Mode

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The difference between expected value of statistic and parameter is called:

- ▶ Non-sampling error
- ▶ Sampling error
- ▶ Standard error
- ▶ **Bias**

In Statistics, we have MSE which is abbreviation of.....

- ▶ **Mean square error**
- ▶ Measured square error
- ▶ Medical screening exam
- ▶ Major sampling error

The F-distribution has parameter.

- ▶ One
- ▶ No
- ▶ **Two**
- ▶ Three

The degrees of freedom for a t-test with sample size 10 is:

- ▶ 5
- ▶ 8
- ▶ **9**
- ▶ 10

The degrees of freedom for a t-test with sample size 14 is:

- ▶ 14
- ▶ **13**
- ▶ 7
- ▶ 0

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The degrees of freedom for a t-test with sample size 6 is:

- ► 6
- ► 3
- ► 5
- ► 7

The difference between statistic and parameter is called:

- ► Bias
- ► Standard error
- ► Sampling error
- ► None of above

In testing of hypothesis, we always begin it with assuming that:

- ► Null hypothesis is true
- ► Alternative hypothesis is true
- ► Sample size is large
- ► Population is normal

Evaluate: (9-4)!

- ► 362880
- ► 120
- ► 24
- ► 6

Mean deviation is always:

- ► Less than S.D
- ► Greater than S.D
- ► Greater or equal to S.D
- ► Less or equal to S.D

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Which formula represents the probability of the complement of event A:

- ► $1 + P(A)$
- ► $1 - P(A)$ (Page 156)
- ► $P(A)$
- ► $P(A) - 1$

Ideally the width of confidence interval should be:

- ► 0 (Page 270)
- ► 1
- ► 99
- ► 100

A discrete probability function $f(x)$ is always:

- ► Non-negative
- ► Negative
- ► One
- ► Zero

An expected value of a random variable is equal to:

- ► Variance
- ► Mean
- ► Standard deviation
- ► Covariance

The area under a normal curve between 0 and -1.75 is

- ► .0401
- ► .5500
- ► .4599
- ► .9599

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The continuity correction factor is used when:

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Which of the following is impossible in sampling:

- ▶ Destructive tests
- ▶ Heterogeneous
- ▶ To make voters list
- ▶ None of these

Which one of the following statements is true regarding a sample?

- ▶ It is a part of population
- ▶ It must contain at least five observations
- ▶ It refers to descriptive statistics
- ▶ It produces True value

The data for an ogive is found in which distribution?

- ▶ A relative frequency distribution
- ▶ A frequency distribution
- ▶ A joint frequency distribution
- ▶ A cumulative frequency distribution

$10! = \dots\dots\dots$

- ▶ 362880
- ▶ 3628800
- ▶ 362280
- ▶ 362800

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The curve of the F- distribution depends upon:

- ► Degrees of freedom
- ► Sample size
- ► Mean
- ► Variance

When E is an impossible event, then $P\{E\}$ is:

- ► 2
- ► 0
- ► 0.5
- ► 1

Which graph is made by plotting the mid point and frequencies?

- ► Frequency polygon
- ► Ogive
- ► Histogram
- ► Frequency curve

In a set of 20 values all the values are 10, what is the value of median?

- ► 2
- ► 5
- ► 10
- ► 20

A uniform distribution is defined by:

- ► Its largest and smallest value
- ► Smallest value
- ► Largest value
- ► Mid value

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When testing for independence in a contingency table with 3 rows and 4 columns, there are _____ degrees of freedom.

- ▶ 5
- ▶ 6
- ▶ 7
- ▶ 12

When a fair die is rolled, the sample space consist

- ▶ 2 outcomes
- ▶ 6 outcomes
- ▶ 36 outcomes
- ▶ 16 outcomes

The curve of the F- distribution depends upon:

- ▶ Degrees of freedom
- ▶ Sample size
- ▶ Mean
- ▶ Variance

$E(4X + 5) =$ _____

- ▶ $12 E(X)$
- ▶ $4 E(X) + 5$
- ▶ $16 E(X) + 5$
- ▶ $16 E(X)$

In chi-square test of independence the degrees of freedom are:

- ▶ $n - p$
- ▶ $n - p - 1$
- ▶ $n - p - 2$
- ▶ $n - 2$

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In an ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.....

- ► 2, 4
- ► 3, 15
- ► 3, 12
- ► 2, 12

A set that contains all possible outcomes of a system is known as

- ► Finite Set
- ► Infinite Set
- ► Universal Set
- ► No of these

Stem and leaf is more informative when data is :

- ► Equal to 100
- ► Greater Than 100
- ► Less than 100
- ► In all situations

A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ► Infinite population
- ► Finite population
- ► Concrete population
- ► Hypothetical population

For a particular data the value of Pearson's coefficient of skewness is greater than zero. What will be the shape of distribution?

- ► Negatively skewed
- ► J-shaped
- ► Symmetrical
- ► Positively skewed

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In measures of relative dispersion unit of measurement is:

- ► Changed
- ► **Vanish**
- ► Does not changed
- ► Dependent

The F-distribution always ranges from:

- ► 0 to 1
- ► 0 to $-\infty$
- ► $-\infty$ to $+\infty$
- ► **0 to $+\infty$**

The location of the critical region depends upon:

- ► Null hypothesis
- ► **Alternative hypothesis**
- ► Value of alpha
- ► Value of test-statistic

In chi-square test of independence the degrees of freedom are:

- ► **$n - p$**
- ► $n - p - 1$
- ► $n - p - 2$
- ► $n - 2$

The t-Distribution is..... Spread out then the standard normal Distribution.

- ► Less
- ► **More**
- ► Equally
- ► Not

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To find the confidence interval for the ratio of two variances we use:

- ► F-Distribution
- ► Z-Distribution
- ► Chi-Square Distribution
- ► T-Distribution

How many percent of values are less than 4th deciles in a symmetric distribution.

- ► 14
- ► 24
- ► 4
- ► 40

The combined distribution of more than two random variables is:

- ► Bivariate Distribution
- ► Marginal Distribution
- ► Joint Distribution
- ► Univariate Distribution

The degrees of freedom for a T-test with sample size 14 is:

- ► 14
- ► 13
- ► 7
- ► 0

Which of the following is true for the binomial distribution $b(x; n, p)$:

- ► Mean > Variance
- ► Mean < Variance
- ► Mean = Variance
- ► Mean = Standard Deviation

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What is m in the formula of mode?

- first frequency
- last frequency
- middle frequency
- highest frequency

The parameters of the binomial distribution $b(x; n, p)$ are:

- x & n
- x & p
- n & p
- x, n & p

Which of the following is true for the Poisson distribution:

- mean > variance
- mean < variance
- mean = variance
- mean = standard deviation

If a significance level of 1% is used rather than 5%, the null hypothesis is:

- More likely to be rejected
- Less likely to be rejected [Click here for detail](#)
- Just as likely to be rejected
- None of the above

The degrees of freedom for a t-test with sample size 10 is

- 5
- 8
- 9
- 10

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The value of χ^2 can never be :

- ► Zero—
- ► Less than 1
- ► Greater than 1
- ► **Negative**

ANOVA was introduced by :

- ► Helmert
- ► Pearson
- ► **R.A Fisher**
- ► Francis

The test statistic used in analysis of variance procedure follow the Distribution.:

- ► χ^2
- ► T
- ► Z
- ► **F**

For testing of hypothesis about population proportion , we use:

- ► **Z-test**
- ► t-Test
- ► Both Z & T-test
- ► F test

A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- ► $\frac{1}{2}$
- ► **$\frac{1}{3}$ $\frac{2}{6} = \frac{1}{3}$**
- ► $\frac{2}{3}$
- ► $\frac{5}{6}$

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The probability of drawing a king of spade from a pack of 52 cards is:

- ► $\frac{1}{4}$
- ► $\frac{1}{13}$
- ► $\frac{1}{26}$
- ► $\frac{1}{52}$

From point estimation, we always get:

- ► Single value
- ► Two values
- ► Range of values
- ► Zero

When c is a constant, then $E(c)$ is:

- ► 0
- ► 1
- ► c
- ► $-c$

The deviation of a distribution from symmetry is called:

- ► Kurtosis
- ► Skewness
- ► Dispersion
- ► Flatness

If E is an impossible event, then $P(E)$ is:

- ► 1
- ► 2
- ► 0
- ► 0.5

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If a data set has the even number of observations, the median :

- ▶ Is the average value of the two middle items
- ▶ Can not be determined
- ▶ must be equal to the mean
- ▶ Is the average value of the two middle items when all items are arranged in ascending order

The height of a student is 60 inches. This is an example of?

- ▶ Continuous data
- ▶ Qualitative data
- ▶ Categorical data
- ▶ Discrete data

In Statistics, we have MSE which is abbreviation of.....

- ▶ Mean square error
- ▶ Measured square error
- ▶ Medical screening exam
- ▶ Major sampling error

Var(4X + 5) = _____

- ▶ 16 Var (X)
- ▶ 16 Var (X) + 5
- ▶ 4 Var (X) + 5
- ▶ 12 Var (X)

The hyper geometric random variable is a(an):

- ▶ Continuous variable
- ▶ Discrete variable
- ▶ Undefined
- ▶ Independent variable

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From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming 'yes' means a success, which of the following is correct?

- ▶ Sample proportion $p=0.33$
- ▶ Sample proportion $p=0.25$ (Page 245)
- ▶ Population proportion $p= 0.33$
- ▶ Population proportion $p=0.2$

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

- ▶ Mutually exclusive
- ▶ Equally likely
- ▶ Not mutually exclusive
- ▶ Exhaustive

The LSD test is applied only if the null hypothesis is:

- ▶ Rejected
- ▶ Accepted
- ▶ No conclusion
- ▶ Acknowledged

Analysis of variance is a procedure that enables us to test the equality of several:

- ▶ Variances
- ▶ Means
- ▶ Proportions
- ▶ Groups

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If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► 0,1,2,3
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

If X and Y are independent variables, then $E(XY)$ is:

- ► $E(XY)$
- ► $E(X) \cdot E(Y)$
- ► $X \cdot E(Y)$
- ► $Y \cdot E(X)$

The parameters of the binomial distribution $b(x; n, p)$ are:

- ► x & n
- ► x & p
- ► n & p
- ► x, n & p

If $P \in$ is the probability that an event will occur, which of the following must be false:

- ► $P \in = -1$
- ► $P \in = 1$
- ► $P \in = 1/2$
- ► $P \in = 1/3$

The analysis of variance technique is a method for :

- ► Comparing F distributions
- ► Comparing three or more means
- ► Measuring sampling error
- ► Comparing variances

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The continuity correction factor is used when:

- ► The sample size is at least 5
- ► Both nP and $n(1-P)$ are at least 30
- ► A continuous distribution is used to approximate a discrete distribution
- ► The standard normal distribution is applied

The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:

- ► Descriptive Statistics
- ► Advance Statistics
- ► Inferential Statistics
- ► Sampled Statistics

Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► Tabulation
- ► Bar chart
- ► Component bar chart

Which one of the following statements is true regarding a population?

- ► It must be a large number of values
- ► It must refer to people
- ► It is a collection of individuals, objects, or measurements
- ► It is small part of whole

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In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- ▶ The regression line will pass through the origin
- ▶ The regression line will pass through the point (0,10).
- ▶ The regression line will pass through the point (0,-10).
- ▶ The slope of the line will also be equal to 0.

A failing student is passed by an examiner is an example of:

- ▶ Type I error
- ▶ Type II error
- ▶ Correct decision
- ▶ No information regarding student exams

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مزید ایسی پوسٹ حاصل کرنے کیلئے ہمارا یوٹیوب چینل سبسکرائب کریں
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